1. Beechcroft	5.0km		60	✓	18
2. Colebrook	2.2km		26	✓	8
3. Covingham	2.4km		29	✓	8
4. Eldene	5.4km		65		19
5. Goddard Park	4.3km		52	✓	15
6. Grange	1.8 km	✓	22	✓	6
7. Liden	5.0km		60	✓	18
8. Nythe	2.3m		28	✓	8
9. South Marston	1.8km	✓	22	✓	6
Secondary Schools / Further Education					
10. Kingsdown	4.0km		48	✓	14
11. Dorcan College	3.3km		40	✓	12
12. Swindon New College	4.3km		52	✓	15
13. Swindon College	5.0kim		60	✓	18

Figure 6e: Distance to Education Destinations

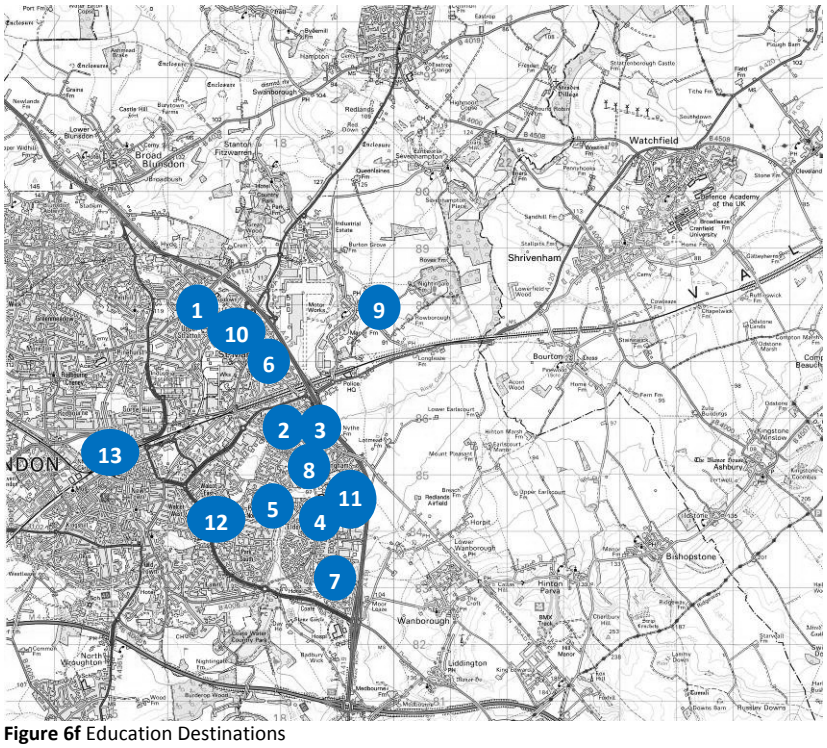


Figure 6f Education Destinations

Miscellaneous Amenities

6.8 A full range of leisure and complementary amenities existing in the area of the development.

6.9 South Marston Village Hall provides a venue for Parish meetings and local community events and is owned and managed by the Parish Council and a local charity. The primary school also provides a playgroup facility for parents and their babies/toddlers. There is a youth club gardening/allotment club, walking club and business club. South Marston also has a church, two public houses and hotel and conference venue. A post office is available nearby as are a range of food and non-food retail outlets.

6.10 South Marston has a recreation area (2.34 Ha) located on the south eastern edge of the village, adjacent to the primary school. The village allotments are located at the northern edge of the village, on the Highworth Road.

6.11 Nightingale Wood, to the east of the site is a general recreation area and County Wildlife Site.

6.12 The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6g, with the location indicated in Figure 6h.

Shops & General Amenity	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
1. Train Station	5.3km		64		19
2. Bank	5.0km		60	✓	18
3. Greenbridge retail park	2.3km		28	✓	8
4. Supermarket	0.5km	✓	6	✓	2
5. Swindon Town centre	5.0km		60	✓	18
6. Post Office	2.3km		28	✓	8
7. South Marston	1.8km	✓	22	✓	6
8. Nightingale Wood	2.0km	✓	24	✓	7

Figure 6g: Distance to Key Local Amenities

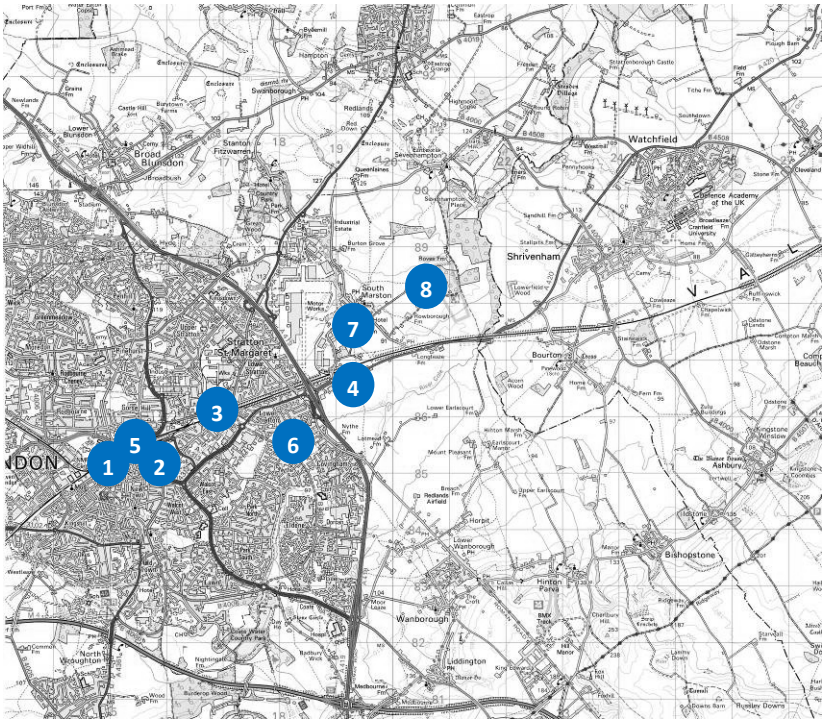


Figure 6h Education Destinations

6.13 It may be concluded that the development will have very good accessibility to a wide range of local amenities that will support the new and existing community. Figure 6i below provides a graphical representation of the 2km walking and 5km cycling isochrones, which the range of local amenities exist. The proposed development will not create any new accessibility barriers within the surrounding area. The range of facilities and services, including the provision made for education will also significantly improve as a result of the application proposals. A higher level of services and facilities

and employment again will be provided within the wider Eastern Villages development most notably the District centre, employment and secondary education provision to the south of the A420 which would be within a 2 km walking isochrone of the site.

Figure 6i: Walking and Cycling Isochrones

Safety

6.14 With new developments comes the potential for increased risk of accidents in the immediate area, due to increased multi-modal traffic. The safety of the development is therefore achieved firstly by identifying the existing accident records and making changes as necessary to the highway network to mitigate any problems. Thereafter, the proposals must be designed to appropriate standards with safety reviews being conducted as necessary during the process. In respect of these requirements:

1. A review of the historical accidents has been completed that confirms there is no accident trend or risk that might materially be increased through the delivery of the development.

2. The proposals have been developed in line with recognised standards in the form of the Design Manual for Roads and Bridges and Manual for Streets. A Stage One Road Safety Audit will be completed with regards to the site access to ensure compliance with the relevant applied design standards.

Economy

6.15 The transport economic efficiency of the development is achieved in part through the successful delivery of a comprehensive transport access strategy that considers all modes of transport, to ensure journey reliability.

6.16 A key transport economic objective will be to minimise any detrimental impact upon private road users' journey time, reliability and travel costs, and to maintain or reduce public transport and non-motorised journey times. The former will be achieved through assessment of trip generation and its likely impact on the local road network, and the latter through a considered choice of mitigating measures; both will be discussed in a later section of this Transport Assessment.

6.17 It should be noted that the proposed development will deliver high quality housing in an area of Swindon that will increase the work force to maximise employment opportunities.

Environment

6.18 The transport environmental benefits of the development are achieved in part through the delivery of a sustainable transport strategy that encourages travel by walking, cycling and public transport and reduces the reliance of the single occupancy vehicle trip.

6.19 As a result of development proposals, local traffic generally increase and appropriate mitigation may be implemented to accommodate this effect. Later Chapters in this report highlight that in this case no such mitigation is required. No significant issues are apparent in relation to the environmental issues.

6.20 The development proposal is in line with transport planning policy. The Transport Assessment underlines areas in which the proposal supports local, regional and national planning transport policies as detailed in Chapter 3.

6.21 There will be no exacerbation of social exclusion resulting from the residential development since no existing travel movements will be cut off or hampered.

Summary of Site Accessibility

6.22 This section of the TA demonstrates that the proposed development site has a wide range of locational advantages in terms of site accessibility.

6.23 The site is in close proximity to Swindon and South Marston, which provide a range of amenities to serve daily needs both in relation to food retail and educational needs. The development will reinforce these services as necessary with the provision of a local centre, schools and health facilities. A higher level of services and facilities and employment again will be provided within the wider Eastern Villages development most notably the District centre, employment and secondary education provision to the south of the A420 which would be within a 2 km walking isochrone of the site.

6.24 The development site will have excellent walking and cycling links into Swindon. Future residents will readily be able to access both road and rail public transport networks.

7 Travel Plan

Travel Plan Benefits

7.1 A Travel Plan (TP) is a management tool designed to enable the users of any site to make more informed decisions about their travel while minimising the adverse impacts of the development on the environment. This is achieved by setting out a strategy for eliminating the barriers keeping users of the site from using sustainable modes and managing single-occupancy car use.

7.2 This Transport Assessment is accompanied by a standalone Framework Travel Plan (FTP). The Transport Assessment should be read in conjunction with the FTP to fully understand the overall transport strategy for the site. The FTP has formed the basis for the sustainable transport strategy and FTP for the proposed development. The FTP will establish the overarching principles to be applied to ensure that the final TP will maximise modal shift. The TP was produced in discussion with the HA and SBC.

7.3 A summary of the Travel Plan is provided below.

7.4 If well-designed and properly managed, the implementation of a Travel Plan can lead to a decrease in the proportion of site users using private cars and an increase in the proportion using sustainable modes of public transport, including walking and cycling.

7.5 Travel Plans can also:

- Improve the environmental credentials of the proposed development
- Reduce the traffic impact on the local highway network
- Improve the health and well-being of all the site occupiers
- Reduce adverse impacts on local residents and businesses

Travel Plan Objectives

7.6 The Travel Plan for the development has several key objectives, as listed below:

- Reduction in the car based trips
- Reduction of Single Occupancy Vehicle (SOV)
- Reduction in congestion and pollution through reduced car use
- Improve the modal split of trips made by walking
- Improve the modal split of trips made by cycling
- Improve the modal split of trips made by public transport

Travel Plan Targets

7.7 The TP will need to establish mode share targets. These targets should be based on challenging, but achievable non-car and Single Occupancy Vehicle (SOV) mode share targets. The targets should be based upon current practice in the hinterland around, and the location of, the site. The target will take account of the local geography and existing transport provision.

7.8 Based on the census information, the nature of the development and the expected travel characteristics, the target for the site will be to reduce the SOV by an ambitious 10% modal shift.

Travel Plan Coordinator

7.9 Research has shown that Travel Plans need to be managed by a Travel Plan Coordinator, who has a clear brief with dedicated resources to manage the Plan to ensure that's its objectives are met.

7.10 The Travel Coordinator role will be funded for a period of 5 years.

7.11 The key responsibilities undertaken by the Coordinator as set out below will be reviewed and amended on a regular basis:

- Leading on the delivery of the TP once approved
- Coordinating the necessary data collection required to develop the 'Household Travel Pack' and 'Employee Travel Packs'
- Representing the 'human face' of the TP including liaison with residents' steering group or management committee
- Promoting the individual measures and packages
- Liaising with the relevant Council Public Transport Team
- Liaising with the Local Highway Authority over monitoring and reviews of the TP

- Assessing progress towards achieving mode-shift away from car use

Travel Plan Measures

7.12 The key to a successful TP is identifying the correct measures that will suit the future residents. It is unlikely that there will be sufficient attraction to a single measure and hence a combination of measures is the most appropriate approach to take.

7.13 In order to maximise the uptake of sustainable transport measures of the development, sustainable transport modes will be available and will be promoted during all stages of the development process from the design, construction and initial marketing of the development through to initial occupation and then on to full occupation of the site.

7.14 The key stages of the Travel Plan process integrate with the key stages of the development process as set out below:

- Before occupation of any dwellings - pre occupation
- During the period when dwellings are being occupied – during occupation
- After dwellings have been occupied – post occupation

8 Development Traffic Generation

Introduction

8.1 In the context of the Eastern Villages proposals, the primary objective of transport network modelling is to provide the tool to assess the effects of growth on the transport network and help inform the need for interventions to ensure the network operates satisfactorily into the future. Scoping discussions with representatives of the HA and SBC concluded that the impact of the development should be considered and appraised using the existing traffic network models for Swindon. These include the Saturn model identified as the Swindon Multi-Model Transport Model (SMMTM) and the A419/A420 Paramics strategic model.

8.2 The SMMTSM is a validated and calibrated model for a base year of 2006 and covers the geographical area of Swindon. This is a Saturn model, which is suitable for high level strategic modelling. The A419 / A420 Paramics model is a micro-simulation model has been calibrated and validated to a more current base year.

8.3 It was agreed with representatives of the HA and SBC that the A419 / A420 Paramics provides the most appropriate model to test the effects of the development. However, as the SMMTSM model covered a wider geographical area, it was also agreed initially to model development impacts using Saturn, with the results then being used to facilitate the Paramics model. This methodology provided the following advantages:

- The Paramics model can inherit all growth in traffic demand and committed development traffic including trip distributions across the network.

- The SATURN model enables the understanding of the wider reassignment impacts of background traffic that might arise from a large development such as the EV. These are then inherited by the Paramics model.

- The SATURN model enables the understanding of the wider impacts on the local road network, west of the A419, which the Paramics model does not cover.

- Where appropriate, the SATURN model could be used to establish trip distribution from the site for EV trips, which is then inherited by the Paramics model.

- This methodology retains consistency with all previous studies undertaken regarding the EV/EDA development by JMP using the Paramics model.

Assessment Scenarios

8.4 The following development / network scenarios were agreed with representatives of SBC and the HA:

- 2006 base year
- 2026 future year plus committed developments
- 2026 future year plus committed developments plus Phase I*
- 2026 future year plus committed developments plus Phase I plus Phase IA*
- 2026 future year plus committed developments plus Phase I plus Phase IA Plus Phase II*

**** Note:*** The Phase I, IA and II development scenarios are outlined in Section 2 of this report.

Committed developments

8.5 Committed development assumptions to be included to create the 2026 future year from the 2006 base model have been agreed with representatives of SBC and the HA. The assumptions have been entered into the SMMTM variable demand model. The proposed development of 140 houses at the Crown Timber site in South Marston has also been added to the model to the Do Minimum SATURN highway model.

8.6 The introduction of the committed developments resulted in the inclusion of 18,147 houses, 10,938 jobs and 127,500 sqm of retail floor space.

8.7 The proposed development of 140 houses at the Crown Timber site in South Marston has also been added to the model to the Do Minimum SATURN highway model.

8.8 The list of committed developments is included within the Halcrow modelling report, attached in Appendix F.

Committed Infrastructure Schemes

8.9 The following network changes have been made to the SATURN highway model for the forecast year, in accordance with advice from SBC, to create the Do Minimum highway scenario:

- Blunsdon Bypass.
- Commonhead Flyover.
- Schemes associated with Wichelstowe, including the proposed new layout at M4 Junction 16, the signalisation of Croft Road/Wichelstowe Access and Croft Road/Pipers Way and the internal Wichelstowe road network.
- Changes associated with the Union Square development.
- Transfer Bridges and Bruce Street Bridges Great Western Way changes.
- The consented highway improvements associated with the Commonhead mixed development, including the proposed changes at M4 Junction 15, A419 and development access.
- Removal of the following car parks prior to 2016:
  - Cheltenham Street (102)
  - College (98)

- Regents Close (12)
- Princes Street (119)
- Carlton (590)
- Catherine Street (6)
- Queen Street (8)
- Queenstown (145)

- The incorporation of the following cars parks:
  - Union Square: 850 MSCP (short and long stay) and 694 residential and office use
  - Regent's Circus: 238 for food store and 212 for office and leisure
  - 1,000 spaces north of the railway line (long stay).
- The provision of bus services to serve the Wichelstowe development;
- Coding bus service 15A to the Northern Development Area (NDA) and extension into the Tadpole Farm development.
- Removal of the P&R bus service to Groundwell P&R.

Trip Generation

8.10 Trip generation for the proposed development has been agreed between BCL and representatives of SBC / HA by way of a Technical Note reference 203064 contained in Appendix F. The following tables summarise the agreed development trips. For Phase I, trips are shown in Figure 8a.

Trips	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Phase I trips	226	788	833	477

Figure 8a: Trip generation – Phase I

8.11 Trips for the agreed development profile for Phase IA have been developed and agreed with representatives of SBC / HA using TRICS. For Phase IA, trips are shown in Figure 8b.

Trips	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Phase IA trips	460	189	157	415

Figure 8b: Trip generation – Phase IA

8.12 In relation to Phase II, a similar methodology as agreed for Phase I has been applied, with the resultant trips indicated below.

Trips	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Phase II trips	281	1534	1678	898

Figure 8c: Trip generation – Phase II

Distribution

8.13 The trip distribution has been determined, employing the same methodology as the historical work carried out on the EV development by Halcrow and JMP, by using model zone 68, which is a residential area in Dorcan. The resultant trip distribution is shown below.

Destination	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Swindon	74%	65%	65%	74%
A420 (E)	11%	9%	9%	5%
M4 (W)	8%	16%	13%	11%
M4 (E)	4%	8%	12%	6%
A419 (N)	3%	2%	2%	4%

Figure 8d: Trip distribution Phase I

Paramics Modelling

8.14 JMP has previously developed a Paramics microsimulation model of the A419 corridor covering the section between M4 J15 and the Blunsdon Bypass, and also part of the A420, including the junctions with Thornhill Rd and Old Vicarage Lane.

8.15 The SMMTM flows from Saturn were used as input data for the Paramics modelling, which then included the following stages.

Task 1: Base Year Model Results

8.16 Outputs from the existing base year Paramics model (2008) were extracted in the form of link flows, queuing at critical junctions and journey times.

Task 2: Future Year Model – 2026 Do Nothing

8.17 The SATURN model results were reviewed and any necessary changes in terms of committed development or committed network schemes were made to the model.

8.18 The Paramics model was then reviewed in terms of network structure to reflect any changes in committed network schemes. Demand matrices for Paramics were derived from the 2026 Do Nothing (DN) SATURN model. Traffic signals were re-optimised to ensure the smooth movement of traffic in the network and to reduce queuing.

Task 3: Future Year Model – 2026 Do Something – Phase I

8.19 The details of the Phase 1 EV were incorporated into Paramics. The Paramics demand matrices for this option were derived from the SATURN model and traffic signals will be re-optimised.

Task 4: Future Year Model – 2026 Do Something – Phase IA

8.20 Sensitivity tests were performed to analyse the impact of the potential extension to the Sainsbury's site and the delivery of the employment land to the south of the A420.

Task 5: Future Year Model – 2026 Do Something Phase II

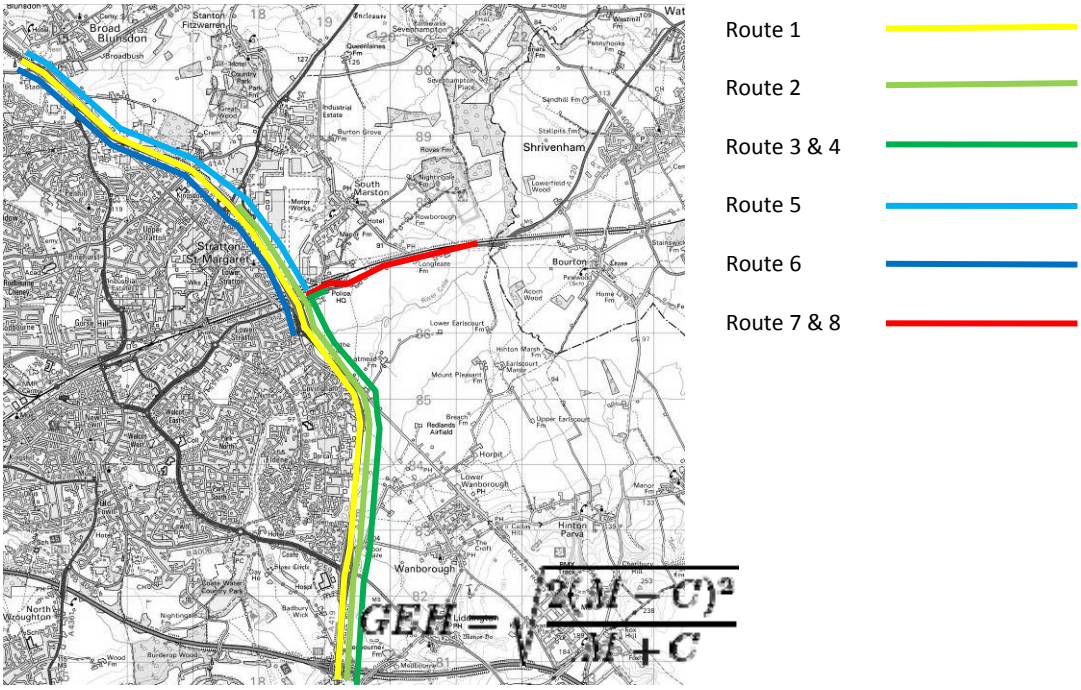
8.21 The details of the Phase II EV development (including quantum of development, location and internal network structure) as well as proposed transport interventions were incorporated into the Paramics model. As before, the Paramics demand matrices for this option were derived from the SATURN model and traffic signals will be re-optimised.

9 Local Road Network Review – Journey Times

Journey Time Assessment Introduction

9.1 The Paramics model has been used to provide estimation of journey times across key routes through the A419 and A420 corridors. The impact on average journey times can be compared between the 'with' and 'without' development scenario tests to identify the impacts of development. The agreed routes that are applicable to this development are indicated below:

- **Route 1:** A149 southbound from Blunsdon Bypass to south of M4 J15
- **Route 2:** A419 northbound from south of M4 J15 to Rattrap Junction
- **Route 3:** A419 northbound from south of M4 J15 to Gable Cross A420
- **Route 4:** A419 southbound from Gable Cross A420 to south of M4 J15
- **Route 5:** A419 southbound from Blunsdon Bypass to White Hart Off-slip
- **Route 6:** A419 northbound from Merlin Way Junction to Blunsdon Bypass
- **Route 7:** A420 eastbound from White Hart to east of Old Vicarage Lane
- **Route 8:** A420 westbound from east of Old Vicarage Lane to White Hart Junction



$$GEH = \sqrt{\frac{(M-C)^2}{M+C}}$$

Figure 9a: Journey Time results

9.2 To assess the significance of the difference in flows and journey times, the Design Manual for Roads and Bridges at Volume 12, Section 2 outlines a methodology known as GEH Statistic, which are frequently employed to compare to sets of model data. The formula for the GEH statistic is indicated below:

Where M and C represents the two sets of data

9.3 As an initial appraisal, the GEH statistics are used to appraise the increase in journey times as a result of the development generated trips. Where this assessment shows a GEH of less than 5, this indicates no material impact.

Phase I Journey Time Results

9.4 The results of the journey time assessment are indicated in the figures below.

Route	2026	2026 with Phase I	GEH
Route 1 –AM	502	575	3.1
Route 1 - PM	548	578	1.2
Route 2 –AM	553	594	1.7
Route 2 - PM	726	732	0.2
Route 3 - AM	588	603	0.6
Route 3 - PM	784	794	0.3
Route 4 - AM	406	503	4.5
Route 4 - PM	499	435	3.0
Route 5 - AM	176	172	0.3
Route 5 - PM	223	258	2.2
Route 6 - AM	188	228	2.8
Route 6 - PM	427	390	1.9
Route 7 - AM	167	250	5.7
Route 7 - PM	199	263	4.2
Route 8 - AM	183	239	3.9
Route 8 - PM	477	369	5.3

Figure 9b: Phase I Journey Time Results, [seconds]

9.5 This comparison of journey times indicates that in morning peak, the majority of the results predict a GEH of less than of 5, therefore demonstrating no material impact. Only on a single Route is there a GEH in excess of 5, on Route 7 A420 eastbound from White Hart to east of Old Vicarage Lane. This section is a major route into the development and as such will experience the highest increase in traffic thus the result is not unexpected. The predicted GEH is reported to be 5.7, which is marginally above 5. Considered in the context of the wider proposals and present / future mitigation along this corridor, this singular impact is not considered to be unacceptable.

9.6 In the evening peak only one Route reports a GEH of greater than 5, being Route 8 the A420 westbound from east of Old Vicarage Lane to White Hart Junction. A GEH is reported at 5.3, but this is a beneficial result as this represents a reduction in journey time between the base and Phase I proposals, from 477 seconds to 369 seconds. This is a result of the interventions proposed at the Gablecross and White Hart interchanges.

9.7 A further beneficial effect is reported along Route 4, the A419 southbound from Gable Cross A420 to south of M4 J15, with a reduction from 499 seconds to 435 seconds. This is a result of the interventions proposed at the Gablecross and White Hart interchanges.

9.8 Overall, it may be concluded that delivery of Phase I will not significantly effect the operation of the road network.

Phase IA Journey Time Results

9.9 The results of journey time assessments are indicated figures below. This comparison assumes that Phase IA would be

approved subsequent to Phase I, therefore becoming a consented scheme forming the base transport environment for comparative reasons. The Phase I, transport interventions as outlined in this report have been tested in this scenario.

9.10 This test indicates that three routes have predicted increases in journey time that give rise to a GEH in excess of five, viz:

- Route 1 – A149 southbound from Blunsdon Bypass to south of M4 J15
- Route 4 - A419 southbound from Gable Cross A420 to south of M4 J15
- Route 8 - A420 westbound from east of Old Vicarage Lane to White Hart Junction

9.11 The GEH for Route 1 is predicted to be 5.6 in the evening peak, with a GEH of 6.0 for Route 4 in the evening peak and a GEH of 9.2 for Route 8 in the morning peak period. These results indicate there will be increases in journey times for traffic travelling southbound along the A419 and westbound along the A420. This would indicate that additional mitigation may be required for the impact of the Phase IA development.

Route	2026 with Phase I	2026 with Phase 1A	GEH
Route 1 - AM	575	574	0.1
Route 1 - PM	578	720	5.6
Route 2 - AM	594	611	0.7
Route 2 - PM	732	697	1.3
Route 3 - AM	603	610	0.3
Route 3 - PM	794	788	0.2
Route 4 - AM	503	488	0.7
Route 4 - PM	435	571	6.0
Route 5 - AM	172	173	0.1
Route 5 - PM	258	259	0.1
Route 6 - AM	228	248	1.3
Route 6 - PM	390	350	2.1
Route 7 - AM	250	281	1.9
Route 7 - PM	263	255	0.5
Route 8 - AM	239	405	9.2
Route 8 - PM	369	447	3.9

Figure 9c: Phase IA Journey Time Results, [seconds]

Phase II Journey Time Results

9.12 The results of journey time assessments are indicated below. This comparison assumes that Phase II would be approved subsequent to Phase IA, therefore becoming a consented scheme and will form the base transport environment. The Phase II, wider EV interventions as outlined in the SBC transport strategy have been tested in this scenario.

9.13 This test indicates that all the routes except Route 2 have a predicted GEH in excess of five. However, several of these are based on beneficial reduction in journey times. The routes reporting a GEH in excess of 5.0 due to journey time increases are identified below:

- Route 3 - A419 northbound from south of M4 J15 to Gable Cross A420

- Route 5 – A419 southbound from Blunsdon Bypass to White Hart Off-slip
- Route 6 – A419 northbound from Merlin Way Junction to Blunsdon Bypass
- Route 8 - A420 westbound from east of Old Vicarage Lane to White Hart Junction

9.14 This would indicate that additional mitigation may be required to mitigate the impact of the Phase II development.

Route	2026 with Phase IA	2026 with Phase II	GEH
Route 1 - AM	574	560	0.6
Route 1 - PM	720	465	10.5
Route 2 - AM	611	628	0.7
Route 2 - PM	697	618	3.1
Route 3 - AM	610	944	12.0
Route 3 - PM	788	885	3.3
Route 4 - AM	488	392	4.6
Route 4 - PM	571	411	7.2
Route 5 - AM	173	290	7.7
Route 5 - PM	259	169	6.1
Route 6 - AM	248	205	2.9
Route 6 - PM	350	481	6.4
Route 7 - AM	281	202	5.0
Route 7 - PM	255	229	1.6
Route 8 - AM	405	929	20.3
Route 8 - PM	447	1163	25.2

Figure 9d: Phase II Journey Time Results, [seconds]

9.15 The development of the Phase I proposals, being the subject of this proposed application, is not predicated on the successful delivery of the Phase II wider EV proposals and interventions forming the SBC strategy for the area as a whole. However, it is important that the Phase I proposals do not compromise the wider growth by the implementation of interventions that might fetter the opportunity for delivery of the intended future infrastructure. This matter is considered in greater detail in Section 10 and 11 of this document.

Journey Time Summary

9.16 Journey time assessments indicate no significant impact on the highway network when the Phase I development scenario is tested alongside the proposed highway interventions. As would be expected, the assessment also indicates that additional mitigation may needed to support the subsequent Phase IA and Phase II development scenarios. The mitigation suggested for Phase I will not prejudice the later phases and this is demonstrated in greater detail in Sections 10 and 11 of this report.

10 Local Road Network Review – Link Flow Assessment

Phase 1 Link Flow Assessment

10.1 The Paramics model has been employed to predict the volume of traffic on the road links within Swindon in the base case and with development scenarios. As an initial appraisal, the GEH statistics are used to appraise the increase in flows as a result of the development generated trips. Where this assessment shows a GEH of less than 5, this indicates no material impact, whereas a GEH of greater than 5 would then require a more detailed assessment on link capacity.

10.2 The link flows are assessed at specific locations as indicated in Figure 10a. The links that connect into Pound Corner are Old Vicarage Lane south of Highworth Road (the southern arm), Thornhill Road west of Highworth Road (western arm) and Highworth Road north of Old Vicarage Lane (north-east arm)

	Morning Peak			Evening Peak		
Location	2026	Phase 1	GEH	2026	Phase 1	GEH
M4 Motorway	7374	7019	4.2	7813	7746	0.8

A419 between J15 and Commonhead	4159	4187	0.4	3837	3817	0.3
A4259 at Commonhead junction	3339	3448	1.9	3200	3248	0.8
Purley Road at Commonhead junction	595	601	0.2	480	492	0.5
Pack Hill at Commonhead junction	210	226	1.1	117	120	0.3
A419 Between Commonhead And Whitehart	5715	5921	2.7	5393	5404	0.1
A420 at Whitehart	2930	3680	13.0	3051	3974	15.6
Ermin Street at Whitehart	703	747	1.6	835	860	0.9
Oxford Road at Whitehart	1440	1566	3.3	1277	1580	8.0
Merlin Way at Whitehart	1850	2090	5.4	2114	2329	4.6
A419 north of Whitehart	6277	6293	0.2	5845	5785	0.8
Thornhill Road north of Gablecross	530	787	10.0	645	937	10.4
A420 east of Gablecross	2136	2723	11.9	2072	2662	12.1
A420 west of Gablecross	2737	3508	13.8	2868	3749	15.3
Supermarket & truck stop access south of Gablecross	541	584	1.8	806	957	5.1
A420 west of Old Vicarage Lane	2000	2556	11.6	1990	2553	11.8
A420 east of Old Vicarage Lane	2081	2251	3.7	2026	2386	7.7
Old Vicarage Lane north A420	106	502	22.7	66	513	26.3
Old Vicarage Lane south of Highworth Road	101	96	0.5	66	144	7.6
Thornhill Road west of Highworth Road	500	359	6.8	391	282	5.9
Highworth Road north of Old Vicarage Lane	512	376	6.5	372	343	1.5
Fleming Way	1510	1599	2.3	1683	1747	1.5
A4312 Queens Drive	3423	3681	4.3	4038	4240	3.1
A313 Ocotal Way	958	1117	4.9	876	966	3.0
A4312 Drakes Way	2346	2483	2.8	2540	2659	2.3
B4006 Swindon Road	2719	2564	3.0	2566	2553	0.3
B4006 Dorcan way	1060	988	2.3	1504	1605	2.6
Marlow Avenue	462	433	1.4	780	849	2.4
A4259 Queens Drive	1745	1552	4.8	2186	2228	0.9
Wanborough Road (east of A419)	613	647	1.4	742	754	0.4
Covingham Drive	1391	1439	1.3	1383	1555	4.5
Link between Honda RB and Hyde Rd	5633	5593	0.5	5041	5018	0.3
Link between Hyde Road and Bypass	5546	5504	0.6	4927	4935	0.1

Figure 10a: Two way flow link analysis – Phase I

10.3 As demonstrated in the Figure 10a above, several links have a predicted GEH in excess of 5, viz:

- A420 at Whitehart in the morning and evening peak
- Oxford Road at Whitehart in the evening peak
- Merlin Way at Whitehart in the morning peak

- Thornhill Road north of Gablecross in the morning and evening peak
- A420 east of Gablecross in the morning and evening peak
- A420 east of Gablecross in the morning and evening peak
- A420 west of Old Vicarage Lane in the morning and evening peak
- A420 east of Old Vicarage Lane in the evening peak
- Old Vicarage Lane north A420 in the morning and evening peak
- Old Vicarage Lane south of Highworth Road in the evening peak
- Thornhill Road west of Highworth Road in the morning and evening peak (decrease in flows)
- Highworth Road north of Old Vicarage Lane in the morning peak (decrease in flows)

10.4 The proposed development will deliver access points within the Thornhill Road, Old Vicarage Lane and A420 corridors and therefore it is expected that increases in traffic flows and the GEH Statistic will be concentrated at these locations. However, the increases in traffic along these links also have to be considered in the context of the proposed intervention strategy along the A420, which are intended to mitigate the effects of the development. In this sense, the journey time assessments are more representative of the whole network operation.

10.5 Two of the locations identified report a GEH of greater than 5, which result from the reduction in vehicles, being at Thornhill Road and Highworth Road at Pound Corner. The decrease in flows at these locations occurs due to the context of the access proposals, viz the realignment of Old Vicarage Lane which encourages trips into the development. Table 10a predicts that the flows approaching Pound Corner could reduce by up to 141 vehicles.

10.6 In summary therefore whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner.

10.7 Outwith the primary corridors providing access to the site, only three corridors report a GEH in excess of 5, viz:

- Oxford Road at White Hart in the evening peak
- Merlin Way at White Hart in the morning peak
- Old Vicarage Lane south of Highworth Road in the evening peak

10.8 The GEH statistic is predicted to be 8.0 within the Oxford Road corridor in the evening peak period. Notwithstanding the GEH statistic, the two way flow is predicted to be 1,580, which is within the theoretical capacity of the highway when assessed in accordance with the DMRB and therefore no specific mitigation is necessary.

10.9 The GEH statistic is predicted to be 5.4 within the Merlin Way corridor in the morning peak period. Notwithstanding the GEH statistic that only marginally exceeds the threshold, the two way flow is predicted to be 2,090, which is within the theoretical capacity of the highway when assessed in accordance with the DMRB and therefore no specific mitigation is necessary.

10.10 The GEH statistic is predicted to be 7.6 within Old Vicarage Lane south of Highworth Road in the evening peak period. Notwithstanding the GEH statistic, the increase is due to the low base levels of traffic, with the two way flow is predicted to be 144, which is well within the theoretical capacity of the highway when assessed in accordance with the DMRB. Therefore, no specific mitigation is necessary.

Phase 1A Link Flow Assessment

10.11 The link flows for the Phase IA scenario are assessed at specific locations as indicated in Figure 10b. This comparison assumes that Phase IA would approved subsequent to Phase I, therefore becoming a consented scheme forming the base line transport environment.

	Morning Peak			Evening Peak		
Location	Phase 1	Phase IA	GEH	Phase 1	Phase IA	GEH
M4 Motorway	7019	7034	0.2	7746	7692	0.6

A419 between J15 and Commonhead	4187	4196	0.1	3817	3800	0.3
A4259 at Commonhead junction	3448	3504	0.9	3248	3136	2.0
Purley Road at Commonhead junction	601	591	0.4	492	444	2.2
Pack Hill at Commonhead junction	226	226	0.0	120	115	0.5
A419 Between Commonhead And Whitehart	5921	5967	0.6	5404	5412	0.1
A420 at Whitehart	3680	3923	3.9	3974	4137	2.6
Ermin Street at Whitehart	747	747	0.0	860	839	0.7
Oxford Road at Whitehart	1566	1707	3.5	1580	1602	0.6
Merlin Way at Whitehart	2090	2130	0.9	2329	2402	1.5
A419 north of Whitehart	6293	6266	0.3	5785	5698	1.1
Thornhill Road north of Gablecross	787	740	1.7	937	886	1.7
A420 east of Gablecross	2723	2973	4.7	2662	2754	1.8
A420 west of Gablecross	3508	3831	5.3	3749	3865	1.9
Supermarket & truck stop access south of Gablecross	584	689	4.2	957	929	0.9
A420 west of Old Vicarage Lane	2556	2416	2.8	2553	2451	2.0
A420 east of Old Vicarage Lane	2251	2157	2.0	2386	2297	1.8
Old Vicarage Lane north A420	502	451	2.3	513	479	1.5
Old Vicarage Lane south of Highworth Road	96	134	3.5	144	126	1.5
Thornhill Road west of Highworth Road	359	361	0.1	282	270	0.7
Highworth Road north of Old Vicarage Lane	376	383	0.4	343	315	1.5
Fleming Way	1599	1630	0.8	1747	1788	1.0
A4312 Queens Drive	3681	3821	2.3	4240	4329	1.4
A313 Ocotal Way	1117	1233	3.4	966	990	0.8
A4312 Drakes Way	2483	2712	4.5	2659	2701	0.8
B4006 Swindon Road	2564	2402	3.3	2553	2551	0.0
B4006 Dorcan way	988	1002	0.4	1605	1620	0.4
Marlow Avenue	433	455	1.0	849	894	1.5
A4259 Queens Drive	1552	1859	7.4	2228	2239	0.2
Wanborough Road (east of A419)	647	648	0.0	754	741	0.5
Covingham Drive	1439	1475	0.9	1555	1673	2.9
Link between Honda RB and Hyde Rd	5593	5575	0.2	5018	5089	1.0
Link between Hyde Road and Bypass	5504	5479	0.3	4935	4980	0.6

Figure 10b: Two way flow link analysis – Phase IA

10.12 Only two links have a predicted GEH in excess of 5, viz:

- A420 west of Gablecross in the morning peak

- A4259 Queens Drive in the morning peak

The GEH statistic is predicted to be 5.3 within A420 corridor. Notwithstanding the GEH statistic is marginally above the threshold, the two way flow is predicted to be 3,831, which is within the theoretical capacity of the highway when assessed in accordance with the DMRB and therefore no specific mitigation is necessary.

As similarly predicted for the Phase I results, there is a reduction in flows on approach to Pound Corner.

10.13 The GEH statistic is predicted to be 7.4 within the Queens Drive corridor in the morning peak period. Notwithstanding the GEH statistic, the two way flow is predicted to be 1,859 when assessed in accordance with the DMRB, which is within the theoretical capacity of the highway and therefore no specific mitigation is necessary.

Phase II Link Flow Assessment

10.14 The link flows for the Phase II scenario are assessed at specific locations as indicated in Figure 10c. This comparison assumes that Phase II would approved subsequent Phase IA, therefore becoming a consented scheme and will form the base line transport environment.

	Morning Peak			Evening Peak		
Location	Phase IA	Phase II	GEH	Phase IA	Phase II	GEH
M4 Motorway	7034	6182	10.5	7692	6556	13.5

A419 between J15 and Commonhead	4196	3771	6.7	3800	3721	1.3
A4259 at Commonhead junction	3504	3077	7.4	3136	2918	4.0
Purley Road at Commonhead junction	591	912	11.7	444	982	20.1
Pack Hill at Commonhead junction	226	330	6.2	115	1042	38.5
A419 Between Commonhead And Whitehart	5967	6138	2.2	5412	5933	6.9
A420 at Whitehart	3923	1755	40.7	4137	2606	26.4
Ermin Street at Whitehart	747	685	2.3	839	699	5.0
Oxford Road at Whitehart	1707	981	19.8	1602	1199	10.8
Merlin Way at Whitehart	2130	761	36.0	2402	797	40.1
A419 north of Whitehart	6266	6477	2.6	5698	6586	11.3
Thornhill Road north of Gablecross	740	1077	11.2	886	1150	8.3
A420 east of Gablecross	2973	2178	15.7	2754	1859	18.6
A420 west of Gablecross	3831	3434	6.6	3865	3512	5.8
Supermarket & truck stop access south of Gablecross	689	378	13.5	929	475	17.1
A420 west of Old Vicarage Lane	2416	1789	13.7	2451	1621	18.4
A420 east of Old Vicarage Lane	2157	1541	14.3	2297	1536	17.4
Old Vicarage Lane north A420	451	356	4.7	479	443	1.7
Old Vicarage Lane south of Highworth Road	134	361	14.4	126	438	18.6
Thornhill Road west of Highworth Road	361	511	7.2	270	545	13.6
Highworth Road north of Old Vicarage Lane	383	382	0.1	315	405	4.7
Fleming Way	1630	1682	1.3	1788	1954	3.8
A4312 Queens Drive	3821	3992	2.7	4329	4602	4.1
A313 Ocotal Way	1233	1345	3.1	990	1228	7.1
A4312 Drakes Way	2712	2822	2.1	2701	2280	8.4
B4006 Swindon Road	2402	2386	0.3	2551	2381	3.4
B4006 Dorcan way	1002	1120	3.6	1620	1585	0.9
Marlow Avenue	455	426	1.4	894	871	0.8
A4259 Queens Drive	1859	1896	0.9	2239	2539	6.1
Covingham Drive	1475	1505	0.8	1673	1655	0.4
Link between Honda RB and Hyde Rd	5575	5609	0.5	5089	5226	1.9
Link between Hyde Road and Bypass	5479	6033	7.3	4980	4870	1.6

Figure 10c: Two way flow link analysis – Phase II

10.15 This demonstrates numerous links that have a predicted GEH in excess of 5, including beneficial results reflecting decreases in link flows. A review of the link flows, where the GEH Statistic is greater than 5, indicates these would be within theoretical highway capacity when assessed in accordance with the DMRB.

10.16 The results also indicate that there could potentially be an increase in traffic through South Marston. As such,

appropriate further mitigation may be needed to support the Phase II scenario at that time taking the form for instance of additional traffic management measures.

Link Flow Summary

10.17 The development proposals will increase traffic in the surrounding area. All links have been tested in accordance with DMRB standards to determine the magnitude of any impacts and where appropriate the highway link capacity. IN the Phase I and IA scenarios, all highways where are material impact is predicted will continue to operate within theoretical highway capacity when assessed in accordance with the DMRB

10.18 Additional mitigation may be needed for the future Phase II scenario, but this will not be prejudiced as a result of the current Phase I proposals.

11 Local Road Network Review – Junction Assessment

Junction Assessment Introduction

11.1 The Paramics model has been used to predict the level of vehicular queuing at the highway junctions across the road network. As an initial appraisal, the GEH statistics are used to appraise the increase in queuing as a result of the development generated trips. Where a junction assessment shows a GEH of less than five, this indicates no material impact, whereas GEH of greater than five would then suggest a more detailed assessment of that junction.

11.2 The reported queue length is the maximum queue on a particular junction approach measured every five minutes during the peak hour, which is then averaged. The maximum queue will be taken to be the longest of the queues on a multi-lane approach, i.e. the same length of queue may not be present in all lanes.

11.3 The Paramics model provides Phase I GEH screening results to identify where a more detailed assessment of junction performance, to include also staging and cycle times, should be carried out. The appropriate tool is the industry standard software, Linsig. This tool has been used to investigate the ability of junctions to accommodate the increase in traffic due to the development where the preliminary Paramics model sufggested the need for more detailed and careful consideration. .

M4 Junction 15

11.4 The current junction is a grade separated junction with the M4 running beneath the circulatory carriageway linking to the A419 and A436, as indicated in Figure 11a below.

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Figure 11a: M4 Junction 15

11.5 The results of the Paramics queue length analysis are indicated in Figure 11b.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
A419 Southbound approach	108	247	10.4	278	230	3.0
M4 westbound offslip	120	122	0.2	274	311	2.2
A346 northbound approach	1425	1399	0.7	1610	1584	0.6
M4 eastbound offslip	229	211	1.2	148	149	0.1

Figure 11b: M4 Junction 15, queue length summary Phase I

11.6 The results indicate there will not be a significant impact on the majority of the arms, with only the A419 southbound approach reporting a GEH in excess of five. Paramics predicts an increase in queuing of 139 metres. This reflects an additional 24 PCU's.

11.7 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH

A419 Southbound approach	247	239	0.5	230	690	21.4
M4 westbound offslip	122	106	1.5	311	295	0.9
A346 northbound approach	1399	1436	1.0	1584	1566	0.5
M4 eastbound offslip	211	237	1.8	149	159	0.8
Phase II Comparison						
outh Arm	**AM Peak**			**PM Peak**		
	2026 with Phase IA	**2026 with Phase II**	**GEH**	**2026 with Phase IA**	**2026 with Phase II**	**GEH**
A419 Southbound approach	239	83	12.2	690	90	30.4
M4 westbound offslip	106	410	18.9	295	228	4.1
A346 northbound approach	1436	1365	1.9	1566	1530	0.9
M4 eastbound offslip	237	268	1.9	159	285	8.4

Figure 11c: M4 Junction 15, queue length summary Phase IA and Phase II

11.8 These results indicate that once Phase IA is included, there is only a single link that predicts a GEH in excess of 5.0. For the Phase II test, only the M4 westbound offslip reports an increase in queuing that results in a GEH in excess of 5.0.

11.9 Improvements to the A419 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP). The Phase 1 development is expected to make contributions toward the IDP, as will the Phase IA and Phase II scenarios, when these are progressed. This being a strategic element of infrastructure delivery, it is proposed that mitigation is supported through infrastructure funding rather than a site specific intervention.

Honda South Roundabout

11.10 This roundabout is located to the east of the A419 and west of the development. The roundabout provides access to and from the southbound A419 and the A361 and provides access to the Honda car manufacturing works. The broad layout of this junction is indicated below.

Figure 11d: Honda South Roundabout

11.11 The results of the Paramics queue length analysis is indicated in Figure 11e.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
Northern approach	11	14	0.9	79	96	1.8
Eastern approach	7	8	0.4	75	105	3.1
Western approach	20	26	1.3	22	20	0.5

Figure 11e: Honda South Roundabout, queue length summary Phase I

11.12 The results indicate there will not be a significant impact on the majority of the arms, all reporting a GEH less than 5.0. The maximum increase is reported on the eastern approach in the evening peak and predicts an increase in queuing of 30 metres. This reflects an additional 5 PCU's. This is not considered a significant increase and the operation of the junction is unlikely to be materially affected.

11.13 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH

Northern approach	14	14	0.1	96	139	3.9
Eastern approach	8	9	0.2	105	90	1.5
Western approach	26	24	0.4	20	21	0.5
Phase II Comparison						
outh Arm	**AM Peak**			**PM Peak**		
	2026 with Phase IA	**2026 with Phase II**	**GEH**	**2026 with Phase IA**	**2026 with Phase II**	**GEH**
Northern approach	14	47	6.0	139	159	1.6
Eastern approach	9	18	2.6	90	28	8.0
Western approach	24	944	41.8	21	42	3.8

Figure 11f: Honda South Roundabout, queue length summary Phase IA and Phase II

11.14 These results indicate that once Phase IA is included, there are no links that are predicted to have an GEH in excess of 5.0. For the Phase II test, only the western approach reports an increase in queuing that results in a GEH in excess of 5.0. This link would require investigation in support of the Phase II scenario to identify the need for appropriate mitigation.

The Rat Trap Roundabout

11.15 The current three arm roundabout is located to the west of the A419 and the proposed development. The roundabout provides access to and from the northbound A419 and the A361, which in turn provides access to the Honda car manufacturing works. The broad layout of this junction is indicted below.

Figure 11g: Rat Trap Roundabout

11.16 The results of the Paramics queue length analysis is indicated in Figure 11h.

Arm	**AM Peak**			**PM Peak**		
	2026 Base	**2026 with Phase I**	**GEH**	**2026 Base**	**2026 with Phase I**	**GEH**
Eastern approach	9	8	0.3	61	56	0.7

Southern approach	72	122	5.1	253	402	8.2
Western approach	381	854	19.0	45	76	4.0

Figure 11h: Rat Trap junction, queue length summary Phase I

11.17 The results indicate that this junction could be affected by the Phase I proposals and a mitigation strategy may be required.

11.18 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Eastern approach	8	11	1.1	56	30	4.0
Southern approach	122	114	0.7	402	394	0.4
Western approach	854	1053	6.4	76	62	1.7
Phase II Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase IA	2026 with Phase II	GEH	2026 with Phase IA	2026 with Phase II	GEH
Eastern approach	11	90	11.2	30	33	0.6
Southern approach	114	1108	40.2	394	1094	25.6
Western approach	1053	447	22.1	62	925	38.9

Figure 11i: Rat Trap junction, queue length summary Phase IA and Phase II

11.19 These results indicate that once Phase IA and more specifically when Phase II is included, most approaches report an increase in queuing that results in a GEH in excess of 5.0. This junction would require investigation at the appropriate time to identify the appropriate mitigation.

11.20 Improvements to the A419 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP). The Rat Trap junction is a recognised constraint on the highway network. The Phase 1 development is expected to make contributions toward the IDP, as will the Phase IA and Phase II scenarios, when these are progressed. This being a strategic element of infrastructure delivery, it is proposed that mitigation is supported through infrastructure funding rather than a site specific intervention.

Commonhead Junction

11.21 Commonhead Junction lies south west of the proposed development. The current layout for the Commonhead junction is indicated below. This junction provides access to and from the A419 to the A4259 Marlborough Road, Pack Hill and Purley Road.

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Figure 11j: Commonhead Junction layout

11.22 The results of the Paramics queue length analysis is indicated in Figure 11k.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
A419 southbound offslip	16	22	1.4	8	8	0.1
Pack Hill approach	11	18	2.0	2	3	0.4
Purley Road approach	21	26	0.9	8	6	0.6
A419 northbound offslip	0	0	n/a	0	0	n/a
Marlborough Road approach	30	40	1.8	44	49	0.8

Figure 11k: Commonhead Junction, queue length summary Phase I

11.23 The results indicate there will not be a significant impact on any of the arms and therefore it is considered that this junction is unlikely to be materially affected by the Phase I scenario.

11.24 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison		
Arm	AM Peak	PM Peak

	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
A419 southbound offslip	22	16	1.4	8	97	12.4
Pack Hill approach	18	17	0.5	3	9	2.7
Purley Road approach	26	20	1.2	6	58	9.0
A419 northbound offslip	0	0	n/a	0	0	n/a
Marlborough Road approach	40	38	0.3	49	130	8.5
Phase II Comparison						
outh Arm	**AM Peak**			**PM Peak**		
	2026 with Phase IA	**2026 with Phase II**	**GEH**	**2026 with Phase IA**	**2026 with Phase II**	**GEH**
A419 southbound offslip	16	175	16.3	97	126	2.7
Pack Hill approach	17	543	31.5	9	414	27.8
Purley Road approach	20	1144	46.6	58	196	12.3
A419 northbound offslip	0	11	4.6	0	42	9.2
Marlborough Road approach	38	638	32.6	130	1600	50.0

Figure 11l: Commonhead Junction t, queue length summary Phase IA and Phase II

11.25 These results indicate that once Phase IA and more specifically when Phase II is included, most approaches reports an increase in queuing that results in a GEH in excess of 5.0. This junction will require investigation at the appropriate time to identify the appropriate mitigation to support the wider growth.

Blunsdon Bypass

11.26 The Paramics modelling has been employed to provide queue lengths for various junction along the Blunsdon Bypass. The queue length results are indicated below.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with	GEH	2026 Base	2026 with	GEH

		Phase I			Phase I	
Cricklade Rd East Left Turn	89	90	0.2	102	129	2.5
Cricklade Rd East Right Turn	102	101	0.1	94	125	3.0
Cricklade Rd North	88	86	0.2	688	687	0.0
Cricklade Rd South	164	159	0.4	177	176	0.0
B4019 North Left Turn	384	384	0.0	178	199	1.5
B4019 North Ahead	16	25	2.0	48	91	5.2
B4019 South	280	266	0.8	274	321	2.7
B4019 Bridge East	13	9	0.9	69	70	0.1
B4019 East	74	74	0.0	74	73	0.0
B4019 Bridge West	62	61	0.1	72	73	0.1
B4019 North Left Turn	37	67	4.2	95	116	2.0
B4019 North Ahead	162	236	5.2	905	961	1.8

Figure 11m: Blunsdon bypass, queue length summary Phase I

11.27 The results indicate that the majority of the junction along this corridor will materially be unaffected by the proposals. Two links are predicted to marginally exceed the 5.0 threshold, viz:

- B4019 North Ahead – increased queue of 74 metres (13 pcu's)
- B4019 North Ahead - increased queue of 43 metres (7 pcu's)

11.28 It is considered that these increases are unlikely to materially affect the operation of the road network.

11.29 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison		
Arm	AM Peak	PM Peak

	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Cricklade Rd East Left Turn	90	134	4.1	129	152	2.0
Cricklade Rd East Right Turn	101	158	5.0	125	176	4.2
Cricklade Rd North	86	92	0.6	687	690	0.1
Cricklade Rd South	159	164	0.4	176	176	0.0
B4019 North Left Turn	384	385	0.1	199	203	0.3
B4019 North Ahead	25	38	2.4	91	68	2.7
B4019 South	266	251	0.9	321	329	0.5
B4019 Bridge East	9	17	2.2	70	70	0.0
B4019 East	74	73	0.0	73	73	0.0
B4019 Bridge West	61	62	0.2	73	74	0.1
B4019 North Left Turn	67	117	5.2	116	193	6.2
B4019 North Ahead	236	275	2.5	961	929	1.1
Phase II Comparison						
outh Arm	**AM Peak**			**PM Peak**		
	2026 with Phase IA	**2026 with Phase II**	**GEH**	**2026 with Phase IA**	**2026 with Phase II**	**GEH**
Cricklade Rd East Left Turn	134	35	10.8	152	286	9.1
Cricklade Rd East Right Turn	158	61	9.2	176	788	27.9
Cricklade Rd North	92	522	24.6	690	693	0.1
Cricklade Rd South	164	179	1.1	176	177	0.1
B4019 North Left Turn	385	388	0.2	203	390	10.9
B4019 North Ahead	38	56	2.7	68	154	8.2
B4019 South	251	86	12.7	329	634	13.9
B4019 Bridge East	17	16	0.4	70	72	0.2
B4019 East	73	76	0.3	73	63	1.2
B4019 Bridge West	62	78	0.7	74	79	0.5
B4019 North Left Turn	117	57	9.6	193	20	16.7
B4019 North Ahead	275	34	6.6	929	416	19.8

Figure 11n: Blunsdon bypass, queue length summary Phase IA and Phase II

These results indicate that the B4019 North Left Turn junction will exceed the threshold once phase IA is considered, with numerous junction affected once the Phase II is included. These junctions will require further investigation at the appropriate time to identify appropriate mitigations to support the wider growth.

White Hart Interchange

11.30 White Hart Junction lies south west of the proposed development on the A420 / A419 corridor. The junction provides access between the A420 and A419 and to Merlin Way and Ermin Street. The current layout for the White Hart junction is indicated below.

Figure 11o: White Hart Interchange

11.31 As part of the identified interventions, a scheme to signalise the eastern side of the interchange has been developed and tested within the Phase I and Phase IA Paramics model. The proposed junction layout is contained in the Appendix C. The queue length results are indicated below.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
Eastern A420 arm	103	113	1.0	361	249	6.4
Merlin Way approach	104	47	6.6	238	150	6.4
Oxford Road approach	25	22	0.5	34	140	11.4
A419 southbound offslip	145	85	5.6	414	766	14.5
A419 northbound offslip	14	38	4.7	18	69	7.6

Figure 11p: White Hart Interchange, queue length summary Phase I

11.32 The results indicate several links exceeding the 5.0 threshold. Several of these are positive reductions with length of queues reducing. Two links in the evening peak report increase in queues namely:

- Oxford Road approach – an increase of 106 metres, equating to 18 PCU's
- A419 southbound offslip – an increase of 352 metres, equivalent to 61 PCU's

11.33 A similar comparison of the Phase IA and Phase II results is indicated below. The Phase IA model includes the same intervention as Phase I, whereas Phase II will include the diamond interchange intervention.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Eastern A420 arm	113	103	1.0	249	208	2.7
Merlin Way approach	47	48	0.2	150	212	4.6
Oxford Road approach	22	31	1.8	140	175	2.8
A419 southbound offslip	85	84	0.1	766	710	2.0
A419 northbound offslip	38	28	1.8	69	47	2.9
Phase II Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase IA	2026 with Phase II	GEH	2026 with Phase IA	2026 with Phase II	GEH
Eastern A420 arm	103	222	9.3	208	145	4.8
Merlin Way approach	48	40	1.3	212	109	8.1
Oxford Road approach	31	281	20.0	175	411	13.8
A419 southbound offslip	84	63	2.5	710	76	32.0
A419 northbound offslip	28	242	18.5	47	286	18.6

Figure 11q: White Hart Interchange, queue length summary Phase IA and Phase II

11.34 The Phase I GEH screening results indicate the need for more detailed consideration of junction performance. Accordingly, a detailed junction performance assessment has been carried out using the industry standard software, Linsig. This tool has been used to investigate the ability of the junction to accommodate the increase in traffic due to the development.

11.35 The output from Linsig refers to the Degree of Saturation, which measures the predicted flow of vehicles against the junction capacity based on the junction geometry. It is normally accepted that a degree of saturation of 100%, indicates that the junction is typically operating at maximum capacity. Due to the inherent day-to-day variability of traffic flows a Degree of Saturation of 90% is seen as acceptable in operational terms for development impact assessments.

11.36 Linsig also reports the expected average queues lengths and average delays that may be expected at a junction. The maximum queue lengths are reported in the junction assessment results as this provides an indication of the efficiency of a junction's performance.

11.37 The results of the Linsig assessment for the Phase I scenario, with the proposed signalisation intervention as shown on drawing 1335/HL/50 in Appendix C, are indicated below.

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
A419 Southbound offslip	63.2%	4	68.4%	4
A420 westbound approach	67.8%	4	68.7%	4
Merlin Way	59.3%	2	78.1%	6
A4312 approach	28.7%	0	47.0%	0
Emin Street approach	20.5%	0	29.4%	0

Figure 11r: Linsig results – White Hart Interchange, queue length summary Phase I

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
A419 Southbound offslip	70.6%	4	61.1%	4
A420 westbound approach	65.5%	6	76.4%	5
Merlin Way	68.1%	4	81.9%	8
A4312 approach	36.2%	0	50.9%	0
Emin Street approach	31.1%	0	31.1%	0

Figure 11s: Linsig results – White Hart Interchange, queue length summary Phase IA

11.38 The results of the detailed Linsig assessment show the proposed intervention will ensure the junction will operate satisfactorily both with Phase I and Phase IA. The subsequent Phase II development scenario will be supported by the IDP Diamond Interchange improvements to the White Hart Junction.

Gablecross Junction

11.39 Gable Cross junction lies south west of the proposed development on the A420 corridor. The junction provides access to and from the A420 from Thornhill Road and Sainsbury's supermarket. The current layout for the Gable Cross junction is indicated below.

Figure 11t: Gable Cross junction

11.40 As part of the identified interventions, a scheme to increase capacity of Gable Cross by widening the approach arms has been identified. This layout has been included within the Phase I and Phase IA Paramics models, with the layout attached in Appendix C. The queue length results are indicated below.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
Eastbound approach	59	66	0.9	141	190	3.8
Retail Park Approach	7	7	0.0	49	52	0.6
Thornhill Road Approach	127	41	9.4	77	34	5.7
Westbound approach	36	112	8.9	151	129	1.8
Sainsbury's Approach	9	13	1.2	123	102	2.0

Figure 11u: Gable Cross, queue length summary Phase I

11.41 The results indicate several links exceeding the 5.0 threshold. Several of these are positive reductions with length of queues reducing. One link in the morning peak reports increase in queues namely:

- A420 westbound approach – an increase of 76 metres, equivalent to 13 PCU's

11.42 A similar comparison of the Phase IA and Phase II results is indicated below. The Phase IA model includes the same intervention as Phase I, whereas Phase II will include the conversion to a signalled controlled crossroads.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Eastbound approach	66	141	7.4	190	187	0.2
Retail Park Approach	7	17	2.7	52	17	6.0
Thornhill Road Approach	41	67	3.5	34	34	0.0
Westbound approach	112	130	1.7	129	136	0.5
Sainsbury's Approach	13	33	4.2	102	112	1.0
Phase II Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase IA	2026 with Phase II	GEH	2026 with Phase IA	2026 with Phase II	GEH
Eastbound approach	141	109	2.8	187	183	0.3
Retail Park Approach	17	18	0.4	17	60	6.9
Thornhill Road Approach	67	104	4.0	34	123	10.0
Westbound approach	130	233	7.7	136	234	7.2
Sainsbury's Approach	33	113	9.3	112	115	0.3

Figure 11v: Gable Cross, queue length summary Phase IA and Phase II

11.43 The results indicate the introduction of Phase IA will not significantly affect the results, with the greatest impact identified on the eastbound approach. The introduction of Phase II will result in GEH statistics greater than 5.0. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

11.44 The Phase I GEH screening results indicate a the need for more detailed consideration of junction performance. Accordingly, a junction performance assessment has been carried out using industry standard software, Arcady. This has been used to investigate the ability of the junction to accommodate the increase in traffic due to the development.

11.45 The output from Arcady refers to the Ratio of Flow to Capacity (RFC), which measures the predicted flow of vehicles against the junction capacity based on the junction geometry. It is normally accepted that an RFC of 1.000, indicates that the junction is typically operating at maximum capacity. Due to the inherent day-to-day variability of traffic flows a RFC of 0.850 is seen as acceptable in operational terms for development impact assessments.

11.46 Arcady also reports the expected average queues lengths and average delays that may be expected at a junction. The maximum queue lengths are reported in the junction assessment results as this provides an indication of the efficiency of a junction's performance.

11.47 The results of the Arcady assessment, based on demand flows, with the proposed interventions as shown on drawing 1335/HL/051 in Appendix C, are indicated below.

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Eastbound approach	0.640	2	0.821	5
Retail Park Approach	0.000	0	0.000	0
Thornhill Road Approach	0.494	1	0.467	1
Westbound approach	0.649	2	0.618	2
Sainsbury's Approach	0.177	0	0.457	1

Figure 11w: Arcady results –Gable Cross , queue length summary Phase I

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Eastbound approach	0.770	3	0.846	6
Retail Park Approach	0.000	0	0.000	0
Thornhill Road Approach	0.554	1	0.477	1
Westbound approach	0.719	3	0.747	3
Sainsbury's Approach	0.244	0	0.712	2

Figure 11x: Arcady results –Gable Cross , queue length summary Phase IA

11.48 The results indicate that the proposed intervention will operate satisfactorily for both the Phase I and Phase IA development scenarios. The subsequent Phase II development scenario will be supported by the IDP A420 corridor improvements.

Police HQ junction

11.49 The Police HQ signals lie due south of the development on the A420 corridor and provides access for the Police HQ to and from the A420. The current layout for the junction is indicated below.

Figure 11y: Police HQ junction

11.50 The queue length results reported by the modelling are indicated below.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
Eastbound approach	44	52	1.0	48	73	3.3
Southbound approach	6	14	2.5	16	14	0.6
Westbound approach	75	164	8.1	399	353	2.4

Figure 11z: Police HQ, queue length summary Phase I

11.51 The results indicate that the eastbound approach is reported to exceed the 5.0 threshold. The queuing increases by 89 metres, equivalent to 15 PCU's.

11.52 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Eastbound approach	52	88	4.3	73	67	0.8
Southbound approach	14	11	0.8	14	17	0.9
Westbound approach	164	202	2.8	353	205	8.9
Phase II Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase IA	2026 with Phase II	GEH	2026 with Phase IA	2026 with Phase II	GEH
Eastbound approach	88	29	7.7	67	38	4.0
Southbound approach	11	71	9.3	17	69	7.9
Westbound approach	202	213	0.7	205	213	0.5

Figure 11aa: Police HQ junction, queue length summary Phase IA and Phase II

11.53 The results indicate the introduction of Phase IA will not significantly affect the results. The introduction of Phase II will result in GEH statistics greater than 5.0. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

11.54 The Phase I GEH screening results indicate the need for more detailed assessment. Accordingly, a detailed junction performance assessment has been carried out using industry standard software, Linsig. This has been used to investigate the ability of the junction to accommodate the increase in traffic due to the development.

11.55 It is possible to mitigate the impact of the development through altering the signal timings. The results of the Linsig assessment, based on demand flows is indicated below.

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Eastbound approach	80.7%	18	87.0%	29
Southbound approach	20.4%	1	45.5%	2
Westbound approach	86.1%	24	77.0%	19

Figure 11ab: Linsig results Police HQ , queue length summary Phase I

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Eastbound approach	89.7%	39	86.4%	29
Southbound approach	45.1%	2	54.0%	3
Westbound approach	85.5%	32	88.6%	33

Figure 11ac: Linsig results Police HQ , queue length summary Phase IA

11.56 The results indicate that the proposed intervention will operate satisfactorily both with Phase I and Phase IA. The subsequent Phase II development scenario will be supported by the IDP A420 corridor improvements.

Old Vicarage Lane

11.57 Old Vicarage lane junction lies to the south of the proposed development and provides access between the A420 and Old Vicarage Lane. The current layout for the junction is indicated below.

Figure 11ad: Old Vicarage Lane junction

11.58 As part of the identified interventions, a scheme to signalise the junction has been identified. This layout has been included within the Phase I and Phase IA Paramics models, with the layout attached in Appendix C. The queue length results are indicated below.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
Eastbound approach	14	159	15.6	15	110	12.1
Southbound approach	0	148	17.2	0	87	13.2
Westbound approach	0	33	8.2	57	59	0.3

Figure 11ae: Old Vicarage Lane, queue length summary Phase I

11.59 The results indicate GEH statistics in excess of 5.0 both in the morning and evening peak periods on Old Vicarage Lane and the A420. This is primarily as a result of introducing signal control in replacement to the present priority arrangement.

11.60 A similar comparison of the Phase IA and Phase II results is indicated below.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Eastbound approach	159	82	7.0	110	80	3.1
Southbound approach	148	158	0.8	87	89	0.2
Westbound approach	33	70	5.1	59	122	6.6
Phase II Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase IA	2026 with Phase II	GEH	2026 with Phase IA	2026 with Phase II	GEH
Eastbound approach	82	72	1.2	80	74	0.7
Southbound approach	158	116	3.6	89	75	1.6
Westbound approach	70	255	14.5	122	80	4.2
Northbound approach	n/a	174	n/a	n/a	68	n/a

Figure 11af: Old Vicarage Lane, queue length summary Phase IA / Phase 2

11.61 The results indicate the introduction of Phase IA will not significantly affect operation of the network, with the westbound approach in the evening peak increasing by 63 metres, an equivalent of 11 pcu's.

11.62 The introduction of Phase II will result in GEH statistics greater than 5.0 on a single arm. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

11.63 The Phase I GEH screening results indicate a the need for more detailed assessment. Accordingly, a detailed junction performance assessment has been carried out using the industry standard software, Linsig. This tool has been used to investigate the ability of the junction to accommodate the increase in traffic due to the development.

11.64 The output from Linsig refers to the Degree of Saturation, which measures the predicted flow of vehicles against the junction capacity based on the junction geometry. It is normally accepted that a degree of saturation of 100%, indicates that the junction is typically operating at maximum capacity. Due to the inherent day-to-day variability of traffic flows a Degree of Saturation of 90% is seen as acceptable in operational terms for development impact assessments.

11.65 The results of the Linsig assessment, based on demand flows, with the proposed signalisation intervention as shown on drawing 1335/HL/054 in Appendix C, are indicated below.

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Eastbound approach	82.9%	26	84.9%	24
Southbound approach	83.7%	12	85.0%	8
Westbound approach	79.5%	25	73.3%	19

Figure 11ag: Linsig results Old Vicarage Lane, queue length summary Phase I

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Eastbound approach	79.7%	11	81.5%	21
Southbound approach	78.0%	22	72.5%	7
Westbound approach	80.6%	24	82.8%	18

Figure 11ah: Linsig results Old Vicarage Lane, queue length summary Phase IA

11.66 The results indicate that the proposed intervention will operate satisfactorily both with Phase I and Phase IA.

Old Vicarage Lane Shuttle Signals

11.67 The existing bridge under the railway at Old Vicarage Lane is narrow, having a width of circa 4.5m. Proposals to signalise the route beneath the bridge have been developed, by providing a shuttle arrangement.

11.68 To test the performance of the proposed shuttle signal arrangement, a detailed junction performance assessment has been carried out using the industry standard software, Linsig. This tool has been used to investigate the ability of the junction to accommodate the increase in traffic due to the development.

11.69 The output from Linsig refers to the Degree of Saturation, which measures the predicted flow of vehicles against the junction capacity based on the junction geometry. It is normally accepted that a degree of saturation of 100%, indicates that the junction is typically operating at maximum capacity. Due to the inherent day-to-day variability of traffic flows a Degree of Saturation of 90% is seen as acceptable in operational terms for development impact assessments.

11.70 The results of the Linsig assessment, based on demand flows, with the proposed signalisation intervention as shown on drawing 1335/HL/057 in Appendix C, are indicated below.

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Southbound approach	64.1%	5	55.1%	3
Northbound approach	31.7%	1	54.3%	3

Figure 11ai: Linsig results Old Vicarage Lane, queue length summary Phase I

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Southbound approach	62.5%	4	52.7%	3
Northbound approach	31.9%	1	52.2%	3

Figure 11aj: Linsig results Old Vicarage Lane, queue length summary Phase IA

11.71 This indicates that the shuttle signals under the rail bridge operate satisfactorily.

Pound Corner

11.72 Pound corner lies in the centre of South Marston and presently provides a sub-standard highway layout, which is

construed by the proximity of properties adjacent to the highway. The current layout for the junction is indicated below.

Figure 11ak: Pound Corner junction

11.73 To test the performance of the junction, a detailed junction performance assessment has been carried out using the industry standard software, Picady. This tool has been used to investigate the ability of the junction to accommodate the increase in traffic due to the development.

11.74 The output from Picady refers to the Ratio of Flow to Capacity (RFC), which measures the predicted flow of vehicles against the junction capacity based on the junction geometry. It is normally accepted that an RFC of 1.000, indicates that the junction is typically operating at maximum capacity. Due to the inherent day-to-day variability of traffic flows a RFC of 0.850 is seen as acceptable in operational terms for development impact assessments.

11.75 The queue length results are indicated below.

Arm	AM Peak			PM Peak		
	2026 Base	2026 with Phase I	GEH	2026 Base	2026 with Phase I	GEH
Highworth Road	0	0	n/a	0	0	n/a
Old Vicarage Lane	12	10	0.9	1	5	2.1
Thornhill Road	5	5	0.3	4	3	0.5

Figure 12al: Pound Corner, queue length summary Phase I

11.76 The results indicate there will not be a significant impact on any of the arms and therefore it is considered that this junction is unlikely to be materially affected by the Phase I scenario. The level of queuing predicted to occur is unlikely to change significantly and will be likely to be largely indiscernible.

11.77 A similar comparison of the Phase IA and Phase II results is indicated below. This indicates that the increase in queuing is again likely to be indiscernible once phase IA is taken into account.

Phase IA Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase I	2026 with Phase IA	GEH	2026 with Phase I	2026 with Phase IA	GEH
Highworth Road	0	0	n/a	0	0	n/a
Old Vicarage Lane	10	8	0.6	5	7	0.8
Thornhill Road	5	5	0.0	3	4	0.2
Phase II Comparison						
Arm	AM Peak			PM Peak		
	2026 with Phase IA	2026 with Phase II	GEH	2026 with Phase IA	2026 with Phase II	GEH
Highworth Road	0	0	n/a	0	0	n/a
Old Vicarage Lane	8	20	3.2	7	29	5.1
Thornhill Road	5	8	1.2	4	11	2.7

Figure 11am: Old Vicarage Lane, queue length summary Phase IA / Phase 2

11.78 The results indicate the introduction of Phase IA or Phase II will not significantly affect the results. As such Pound Corner will not be materially affected by the development proposals. Nonetheless, SBC has investigated a number of minor improvement options for Pound Corner, which are likely to be supported through the IDP.

Greenbridge Junction

11.79 This junction is located to the west of the development site along the A4312. The layout is indicated below.

Figure 11an: Greenbridge junction layout

11.80 This junction is outside the extents of the Paramics model has been assessed using turning flows extracted from the Saturn model. The results are indicated below.

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Slade Drive	0.651	2	0.506	1
B4006 northbound	0.713	3	0.413	1
Drakes Way	0.504	1	0.638	2
Greenbridge approach	0.503	1	0.753	3
B4006 southbound	0.085	0	0.291	0

Figure 11ao: Greenbridge assessment results, Phase I

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Slade Drive	0.630	2	0.508	1
B4006 northbound	0.731	3	0.453	1
Drakes Way	0.500	1	0.617	2
Greenbridge approach	0.629	2	0.754	3
B4006 southbound	0.091	0	0.292	0

Figure 11ap Greenbridge assessment results, Phase IA

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Slade Drive	0.611	2	0.515	1
B4006 northbound	0.764	3	0.448	1
Drakes Way	0.752	4	0.793	4
Greenbridge approach	0.809	4	0.763	3
B4006 southbound	0.116	0	0.349	1

Figure 11aq: Greenbridge assessment results, Phase II

11.81 The results of the Greenbridge assessment indicates that this junction will operate satisfactorily in all the three development scenarios, with a peak RFC reported for Phase II at 0.809.

Eastern A420 Access

11.82 This junction forms the eastern most point of connection of the proposed development. This junction has been assessed using Linsig, with the results indicated below.

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Site Access approach	49.8%	4	31.0%	2
Westbound approach	64.6%	13	73.9%	17
Eastbound approach	48.6%	9	49.7%	9

Figure 11ar: Linsig results A420 eastern access, queue length summary Phase I

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Max Queue	Degree of Saturation	Max Queue
Site Access approach	46.7%	4	30.3%	2
Westbound approach	65.2%	13	73.2%	17
Eastbound approach	47.2%	xx	48.9%	9

Figure 11as: Linsig results A420 eastern access, queue length summary Phase IA

11.83 This indicates that the junction will operate within acceptable thresholds.

Site Access Point – Thornhill Road

11.84 The access strategy is identified above. The development will be accessed from Thornhill Road via a compact roundabout as illustrated below. This access point has been assessed, with the results indicated below.

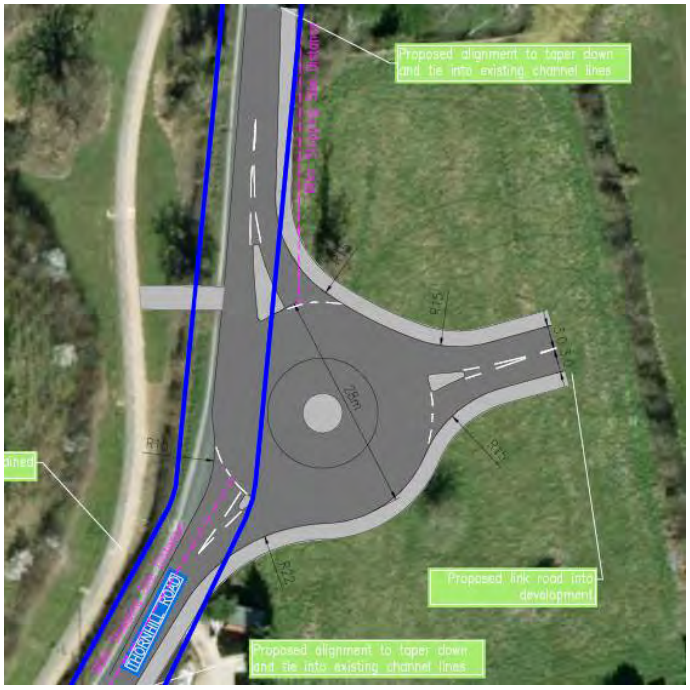


Figure 11at: Thornhill Road Access

Phase I				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Thornhill Road north	0.297	0	0.189	0
Development access	0.275	0	0.153	0
Thornhill Road south	0.290	0	0.614	2
Phase IA				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Thornhill Road north	0.276	0	0.180	0
Development access	0.233	0	0.144	0
Thornhill Road south	0.293	0	0.606	2
Phase II				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Thornhill Road north	0.399	1	0.468	1
Development access	0.765	3	0.510	1
Thornhill Road south	0.333	1	0..787	4

Figure 11au: Thornhill Road access assessment results

11.85 This indicates that the junction operates satisfactorily with all the development scenarios.

Site Access Point – Old Vicarage Lane South

11.86 The development will be accessed from Old Vicarage Lane via a priority junction as illustrated below. This access has been assessed with the results below.

Figure av: Old Vicarage Lane south access

Phase I				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Lane	0.366	1	0.294	0
Development access	0.017	0	0.005	0
Phase IA				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Lane	0.369	1	0.271	0
Development access	0.015	0	0.005	0
Phase II				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Lane	0.209	0	0.194	0
Development access	0.110	0	0.105	0

Figure 11aw: Old Vicarage Lane access assessment results

11.87 This indicates that this junction operates within acceptable limits with a maximum RFC predicted at 0.369.

Site Access Point – Old Vicarage Lane north

11.88 The development will be accessed from Old Vicarage Lane via a roundabout as illustrated below. This access has been assessed with the results below.

Figure ax: Old Vicarage Lane north access

Phase I				
	Morning Peak		Evening Peak	
Links	RFC	Max Queue	RFC	Max Queue
Old Vicarage Road north	0.085	0	0.091	0
Development access east	0.172	0	0.111	0
Old Vicarage Road south	0.059	0	0.045	0
Development access west	0.054	0	0.186	0
Phase IA				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Road north	0.083	0	0.083	0
Development access east	0.154	0	0.103	0
Old Vicarage Road south	0.060	0	0.046	0
Development access west	0.052	0	0.172	0
Phase II				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Road north	0.310	0	0.249	0
Development access east	0.279	0	0.178	0
Old Vicarage Road south	0.149	0	0.171	0
Development access west	0.080	0	0.374	1

Figure 11ay: Old Vicarage Lane roundabout access assessment results,

11.89 This indicates that the will operate within capacity, with a maximum RFC reported at 0.374.

Junction Summary

11.90 The TA has assessed a number of junctions across the Local Road Network and Strategic Road Network that could potentially be affected by the development proposals. Where appropriate, intervention works are promoted to support the development growth promoted in this Phase scenario. Phase IA and Phase II scenarios have also been tested, representing the wider Eastern Villages growth, including the associated interventions.

11.91 The results indicate that the interventions identified are adequate to mitigate the development impact. The junction interventions are identified below.

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road. The layout of this arrangement is shown on drawing 1335-HL-057 and 1335-HL-054 in Appendix C.

- **New Access onto A420:** New signal controlled junction at the east end of phase one to provide access from the A420. The layout of this arrangement is shown on drawing 1335-HL-010 in Appendix C.

- **Police HQ Signals:** Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire.

- **Gablecross** Roundabout: Localised widening on approach arms. The layout of this arrangement is shown on drawing 1335-HL-057 in Appendix C.

- **White Hart interchange:** Partial signalisation of the eastern side of the interchange. The layout of this arrangement is shown on drawing 1335-HL-050 in Appendix C.

11.92 In the immediate environs of South Marston, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included. Whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner. Notwithstanding, the additional traffic generated by the development, with the improvements proposed to the Gablecross roundabout and the junction of Old Vicarage Lane, the level of queuing at Gablecross roundabout on the Thornhill Road arm is forecast to reduce compared to the no development and no improvement position and the queuing on Old Vicarage Lane will be limited and acceptable.

12 Limitations

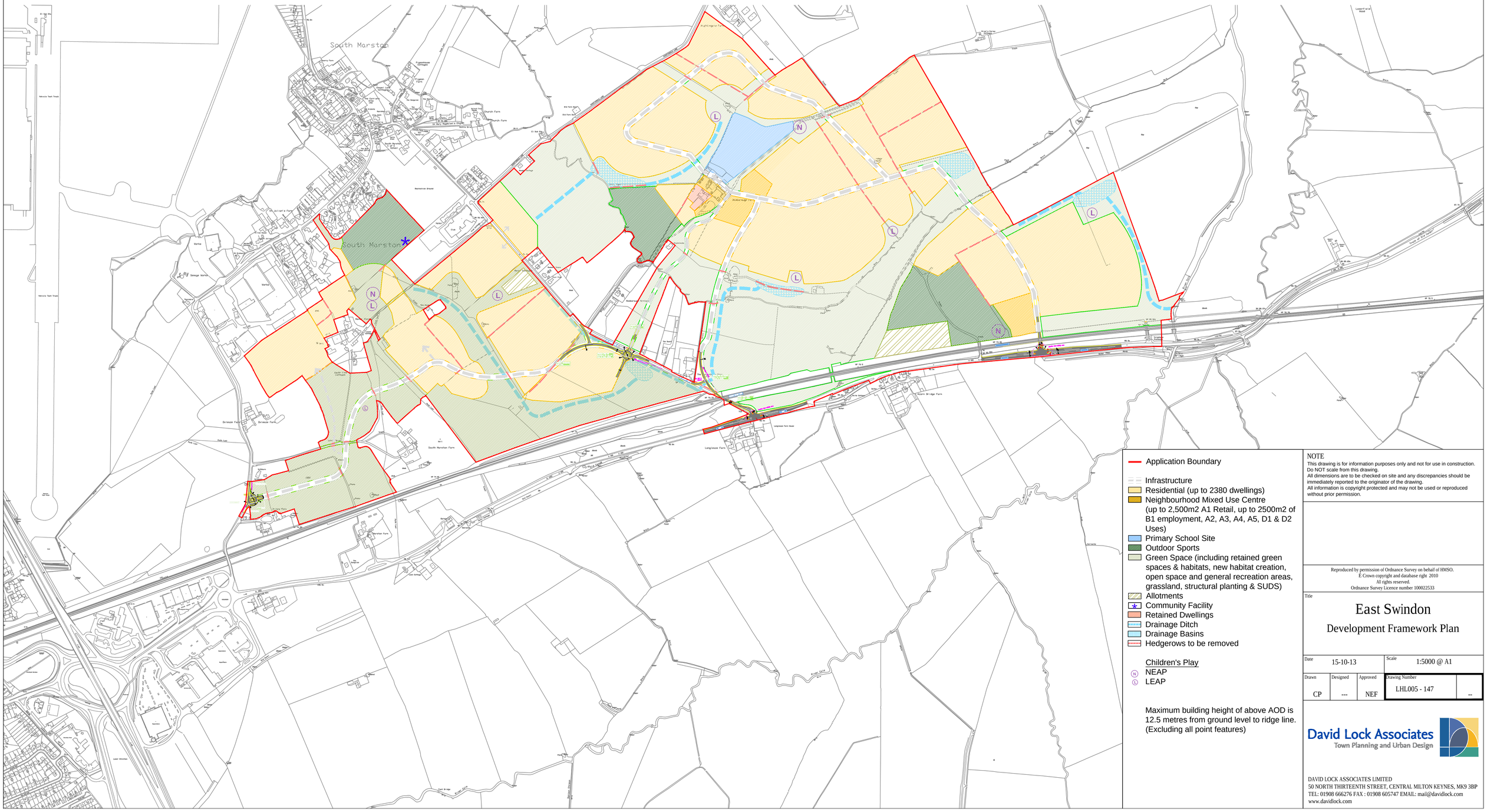
12.1 The conclusions and recommendations highlighted above are limited to the general availability of background information and the planned usage of the site.

12.2 Third party information has been used in the preparation of this report, which Brookbanks Consulting Ltd, by necessity assumes is correct at the time of writing. While all reasonable checks have been made on data sources and the accuracy of data, Brookbanks Consulting Ltd accepts no liability for same.

12.3 The benefits of this report are provided to SEVC for the proposed development land at east of Swindon.

12.4 Brookbanks Consulting Ltd excludes third party rights for the information contained in the report.

Appendix A – Parameter Plan



NOTE
This drawing is for information purposes only and not for use in construction.
Do NOT scale from this drawing.
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Title

East Swindon

Development Framework Plan

Date	Scale
15-10-13	1:5000 @ A1

Drawn	Designed	Approved	Drawing Number	
CP	---	NEF	LHL005 - 147	--

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